

Top Level Testbench Description

Purpose

The SystemVerilog testbench verifies the complete pipelined ALU system through the top-level module. The objective is to validate integration between the control unit, register file, pipeline registers, ALU, and writeback logic.

Clock Generation

The testbench generates a periodic clock used to drive all sequential logic within the design.

Reset Verification

Reset sequences are applied to confirm that internal registers, pipeline stages, and control logic initialize correctly.

Instruction Injection

The testbench applies instruction packets containing opcode and register address information. Instructions are supplied in a controlled sequence to exercise pipeline behavior.

Input Data Loading

External data values are loaded into the register file and used as operands for subsequent ALU operations.

Functional Verification

Each supported ALU operation is exercised and the resulting output values are compared against expected behavior manually.

Flag Verification

Carry, overflow, zero, negative, and comparison flags are monitored to ensure correct generation for each operation.

Pipeline Verification

The testbench verifies that instructions move correctly through all pipeline stages and that writeback occurs after the expected latency.

Waveform Analysis

Simulation waveforms are used to observe internal signals, pipeline registers, addresses, data paths, control signals, and status flags.

Success Criteria

A test passes when the correct result is written to the correct destination register and all associated status flags match expected values.